Tejas Santosh Bhavekar

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Research Interests: Robotics, Mechatronics, Autonomous Robotic Systems, Robotic Hardware & Systems Integration, Perception and Sensing

EDUCATION

- **Indian Institute of Technology Indore**
  B.Tech. in Mechanical Engineering, CGPA: ***8.92/10*** 2023–Present
- **CBSE Board**
  Senior Secondary, ***94.8%*** 2023
- **ICSE Board**
  Secondary, ***94.8%*** 2021

RESEARCH EXPERIENCE

- **Human-Interactive Robotics Lab (HiRo Lab)** *May 2026 - July 2026*
  Research Intern under Prof. Ravi Prakash **Indian Institute of Science, Bangalore**
  – Analyzed Geometry-aware Policy Imitation (GPI) for robot manipulation and identified its policy-averaging failure under homotopy ambiguity through geometric retrieval and vector-field analysis.
  – Developed neural world models for one-step and direct multi-step dynamics prediction, integrating predicted future rewards into GPI for consequence-aware retrieval; achieved 100% success across randomized environments.
  – Implemented closed-loop Panda manipulation in MuJoCo using Mink inverse kinematics, SE(3) task-space control, gripper actuation, trajectory recording, and GPI-based demonstration execution.
- **IASc-INSA-NASI Summer Research Fellowship Program (SRFP)** *May 2025 - July. 2025*
  Research Intern under Dr. Tarun Bangia **ARIES, Nainital**
  – Developed hybrid anomaly detection for azimuth/altitude motors of 3.6m Devasthal Telescope by combining Random Forest, Isolation Forest, One-Class SVM & LSTM Autoencoder with custom time-series features.
  – Developed interactive Streamlit GUI for 3.6m Devasthal Telescope by enabling file-wise anomaly classification, result aggregation, and severity visualization across Normal, Suspicious, and Anomalous labels.
  – Performed FFT-based vibration analysis on large optical assemblies during crane-assisted motion using accelerometer data and validated results with ANSYS modal frequencies to ensure structural safety.

PUBLICATIONS

- **Condition Monitoring Framework for Subsystems of a Ground-Based Telescope** *Oct. 2026*
  Second Author ASME
  – Submitted to the *Journal of Nondestructive Evaluation, Diagnostics and Prognostics of Engineering Systems*; under consideration.

SELECTED PROJECTS

- **Autonomous Rover for Warehouse Rack Inventory Management** *Nov. 2025 - Dec. 2025*
  Inter IIT Tech Meet 14.0 (Problem Statement by Eternal) Drive
  – Led the project of an autonomous rover integrating chassis, motors, sensors, actuators, and controllers.
  – Implemented a belt-driven vertical scanning mechanism using a stepper motor for multi-level shelf scanning.
  – Integrated drive motors, encoders, sensors, and custom wiring harnesses for reliable electromechanical operation.
  – Developed power and safety hardware, including battery monitoring, voltage regulation, and emergency stops.
- **Development of a Device for Characterizing Post-Cooking Traits of Rice** *Aug. 2026 - Present*
  B.Tech Project Under Prof. Ankur Miglani
  – Developed an automated rice characterization system integrating Raspberry Pi, camera, and thermocouples.
  – Implemented computer-vision methods for grain segmentation, geometric measurement, and deformation tracking.
  – Implemented growth-rate-based saturation detection for autonomous soaking-to-cooking stage transition.
  – Designed a closed-loop temperature control system using thermocouple feedback and relay-based heater actuation.

- **3D-Printed Quadruped Robot Design and Development** *Jan. 2026 - Apr. 2026*

  Design of Machine Elements Course Project GitHub

  – Developed a quadruped robot employing a 4-bar linkage mechanism for efficient leg actuation and balance.
  – Led hardware integration involving servo motors, power architecture, system assembly, and overall testing.
  – Implemented crawl and diagonal-pair gait control strategies for stable quadruped locomotion using Arduino.

- **Model CubeSat Project** *Oct. 2024 - Nov. 2024*

  CAE (Computer Aided Engineering) CLUB GitHub

  – Enabled sensing and onboard positioning for the CubeSat using specific environmental and orientation sensors.
  – Designed and assembled a custom PCB to streamline sensor connectivity for accurate data acquisition.

- **Autonomous Lawn Mower** *Aug. 2025 – Present*

  Academic Project under Prof. Vimal Bhatia Drive

  – Designed and assembled a remote-controlled lawnmower using motor drivers, sensors, and embedded hardware.
  – Integrated Raspberry Pi and control circuitry to enable field navigation over defined paths.

- **Micromouse - Maze-solver Robot** *Jul. 2024 - Jan. 2025*

  ROBOTICS CLUB

  – Enabled obstacle avoidance in Micromouse by integrating ultrasonic sensors, IMU, and time of flight sensors.
  – Designed and tested PCB layout to streamline circuitry, improve performance, and make a compact robot.

TECHNICAL SKILLS

- **Hardware & Electronics**: Microcontroller Development (Arduino, ESP32, Raspberry Pi), Sensor integration, PCB Design (Schematic creation, layout design, and testing)
- **Programming**: Python, C++, Java, MATLAB, ROS2
- **Data Analytics**: Feature Engineering, Time-Series Analysis, Predictive & Preventive Maintenance, Anomaly Detection, Machine Learning Fundamentals
- **Design & Simulation**: Fusion 360, AutoCAD, Ansys Software: Static Structural, Modal

RELEVANT COURSEWORK

- **Robotics & Control**: Introduction to Robotics, Instrumentation and Control Systems, Kinematics & Dynamics of Machines, Basics of Automotive Electronics
- **Mechanical Engineering**: Design of Machine Elements, Introduction to Finite Element Methods, Computational Fluids and Structures, Solid Mechanics, Fluid Mechanics, Applied Thermodynamics
- **Mathematics**: Numerical Methods, Linear Algebra, Multivariable Calculus, Differential Equations, Real Analysis, Complex Analysis
- **Electives**: Industrial Data Analytics, Basic Electronic Engineering, Fundamentals of Acoustics, Introduction to Electric Vehicles, Introduction to Hybrid Additive Manufacturing Techniques

LEADERSHIP & SERVICE

- **President**, Robotics Club, IIT Indore *Mar. 2025 - Mar. 2026*
- **Team Lead**, Inter IIT Tech Meet 14.0 *Nov. 2025 - Dec. 2025*
- **Hardware Domain Lead, Core Member**, CAE Club, IIT Indore *Sept. 2024 - Mar. 2026*
- **Organising Team, Robotics Competition**, Robotics Club, IIT Indore *Jan. 2025 - Feb. 2025*

HONORS & AWARDS

- **Bronze Medal**, Inter IIT Tech Meet 14.0, Autonomous Rover for Warehouse Inventory Management. *2025*
- **Bronze Medal**, IITISoC'24, Design & Modelling, Material Selection, Design and FEA of CubeSat *2024*